

PH 113
Physics

Liquid Drop Model

By

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Models of Nuclear Structure

- Why is the binding energy per nucleon almost constant?
- Why do certain nuclei emit α - and β -particles through these particles do not exist inside the nucleus?
- Why are the nuclei containing 2, 8, 20, 28, 50, 82 nucleons most stable?
- Like any other quantum mechanical system, a nucleus also exist in its excited states. The most stable state is the ground state in which the nuclei are generally found.

Models of Nuclear Structure

- The precise nature of the forces acting in the nucleus is unknown.
- Nuclear model are resorted to for investigation and theoretical prediction of its properties.

Models of Nuclear Structure

- Such models may be based on
 - the extrinsic analogy between the properties of atomic nuclei and those of a liquid drop
 - the electron shell of an atoms etc.
- The corresponding models are called the liquid drop model and shell model.

Liquid Drop Model

- The Liquid Drop Model was proposed by the Neils Bohr who observed that there are certain marked similarities between an atomic nucleus and a liquid drop.
- In the Liquid Drop Model, the forces acting in the nucleus are assumed to be analogical to the molecular forces in a droplet of some liquid.

Assumptions

- The nucleus is supposed to be spherical in shape in the stable state, just like as liquid drop is spherical due to the symmetrical surface tension forces.
- The forces of surface tension acts on the surface of the liquid drop. Similarly, there is a potential barrier at the surface of the nucleus.
- The density of a liquid drop is independent of its volume. Similarly, the density of the nucleus is independent of its volume.

Assumptions

- The intermolecular forces in a liquid are short range. Similarly the nuclear forces are short range forces.
- The molecules evaporate from liquid drop on raising the temperature of the liquid due to their increased energy of thermal agitation. Similarly, when energy is given to a nucleus by bombarding it with nuclear projectiles, a compound nucleus is formed which emits nuclear radiations almost immediately.
- When a small drop of liquid is allowed to oscillate, it breaks up into two smaller drops of equal size. The process of nuclear fission is similar and nucleus breaks up into two smaller nuclei.

Semi-Empirical Formula

Using the model of the nucleus as a liquid drop, we attempt to calculate nuclear binding energies.

There are five contributions to the binding energy:

- The volume energy is proportional to the number of nucleons.
- The surface energy is the reduction in the binding energy due to nuclear surface effects.
- The Coulomb energy is the repulsion between protons.

- The asymmetry energy is a reduction in binding energy which occurs when there are more neutrons than protons, and vice versa.
- The pairing energy is somewhat of a "fudge factor" which accounts for stability of even-even and even-odd nuclei, and the reduction of stability for odd-odd nuclei.

Volume Energy

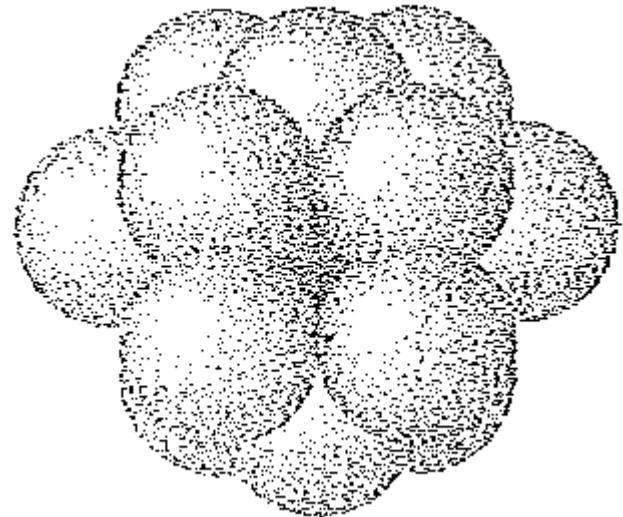
Volume Energy

As a first approximation, we can think of each nucleon in a nucleus as interacting solely with its nearest neighbors.

Energy associated with each nucleon-nucleon bond = U

Because each bond energy is shared by two nucleons therefore each has a binding energy of $\frac{1}{2} U$.

When an assembly of spheres of the same size is packed together into the smallest volume, each interior sphere has 12 other spheres in contact with it



In a tightly packed assembly of identical spheres, each interior sphere is in contact with 12 others

Volume Energy

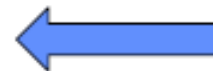
Hence each interior nucleon in a nucleus has a binding energy of $=12(1/2U) = 6U$.

If all A nucleons in a nucleus were in the interior, the total binding Energy would be

$$E_V = 6AU$$

Or

$$E_V = a_1 A$$



Volume energy

Surface Energy

Actually, of course, some nucleons are on the surface of every Nucleus and therefore have fewer than 12 neighbors.

Surface energy $\propto$ surface area

$$\propto 4\pi R^2$$

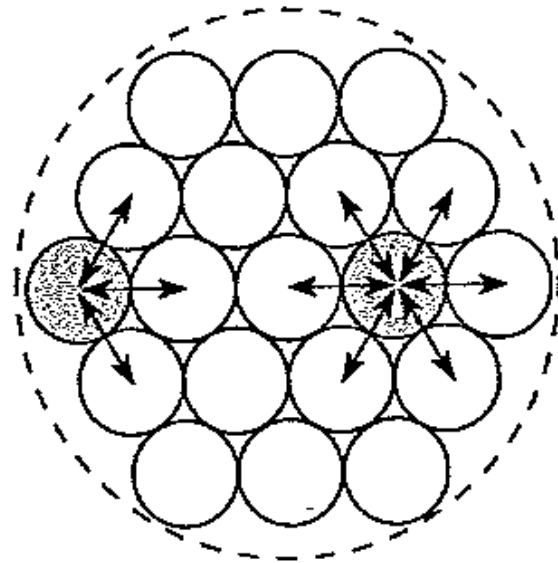
If $R = R_0 A^{1/3}$ then

$$\text{Surface energy} \propto 4\pi R_0^2 A^{2/3}$$

Or

$$\text{Surface energy} = -a_2 A^{2/3}$$

$$\text{Surface energy} \propto A^{2/3}$$



A nucleon at the surface of a nucleus interacts with fewer other nucleons than one in the interior of the nucleus and hence its binding energy is less. The larger the nucleus, the smaller the proportion of nucleons at the surface

Coulomb Energy

The repulsion between each pair of protons in a nucleus also contribute toward decreasing its binding energy.

The coulomb energy E_C of a nucleus is the work that must be done To bring together Z protons from infinity into a spherical aggregate The size of the nucleus.

$$V = -\frac{e^2}{4\pi\epsilon_0 r}$$

Since there are $Z(Z-1)/2$ pairs of protons,

$$E_C = \frac{Z(Z-1)}{2} V = -\frac{Z(Z-1)e^2}{8\pi\epsilon_0} \left(\frac{1}{r} \right)_{av}$$

Coulomb Energy

Where $\left(\frac{1}{r}\right)_{av}$ is the value of $1/r$ averaged over all proton pairs.

If the protons are uniformly distributed throughout a nucleus of radius R , $\left(\frac{1}{r}\right)_{av}$ is proportional to $1/R$ and hence $1/A^{1/3}$

Therefore

$$E_C = -a_3 \frac{Z(Z-1)}{A^{1/3}}$$

The Coulomb energy is negative because it arises from an effect that opposes nuclear stability.

The total binding energy E_b of a nucleus

Surface energies

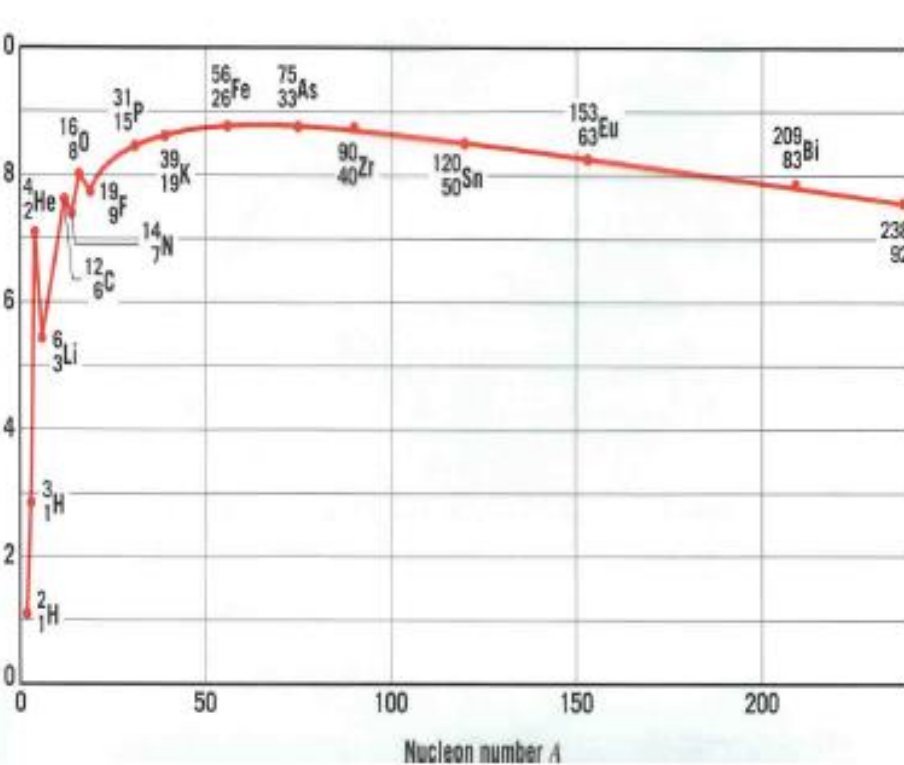
$$E_b = E_v + E_s + E_c = a_1 A - a_2 A^{\frac{2}{3}} - a_3 \frac{Z(Z-1)}{A^{\frac{1}{3}}}$$

Volume energies

Coulomb energies

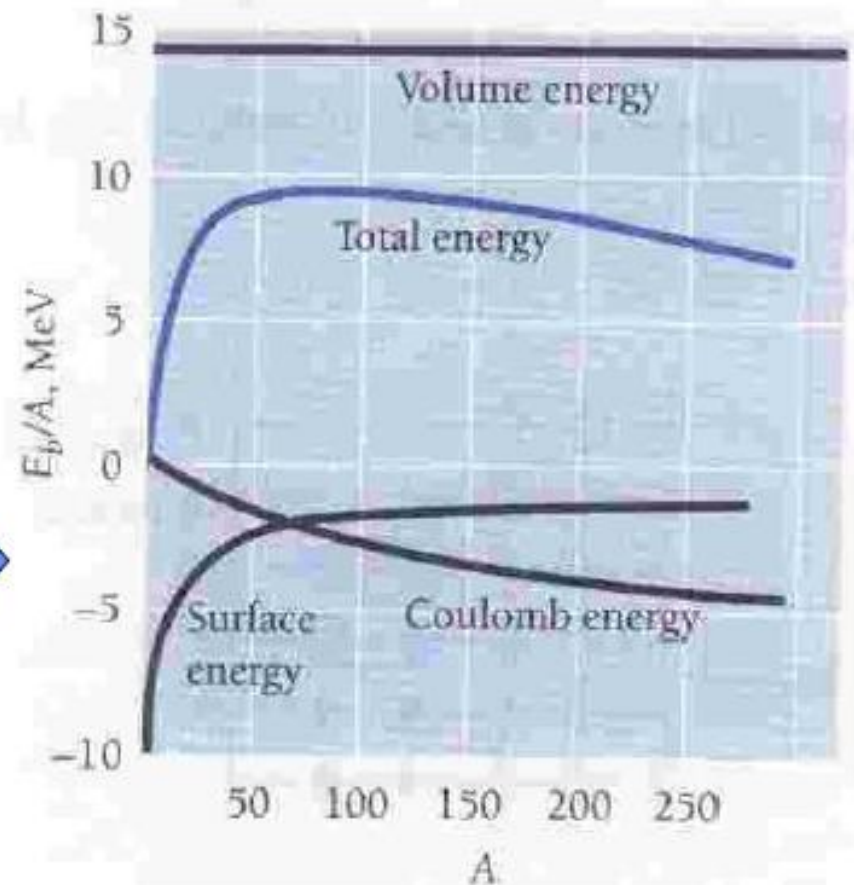
The binding energy per nucleon is

$$\frac{E_{BE}}{A} = a_1 - \frac{a_2}{A^{1/3}} - a_3 \frac{Z(Z-1)}{A^{\frac{4}{3}}}$$



Empirical binding energy per nucleon curve

theoretical binding energy per nucleon curve (using Liquid drop model concept)



Correction to the Formula

The binding energy formula can be improved by taking into account two effects that do not fit into the simple liquid drop model, but which make sense in terms of a model that provides for nuclear energy levels.

Asymmetry Energy

This effect occurs when the neutrons in a nucleus outnumber the protons, which means that higher energy levels have to be occupied than would be the case if N and Z were equal.

Asymmetry Energy

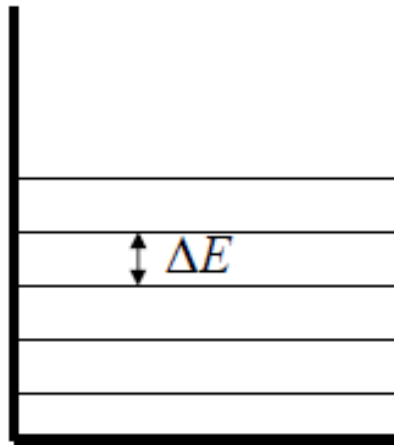
Imagine that the neutrons and protons are in a potential well (the nucleus).

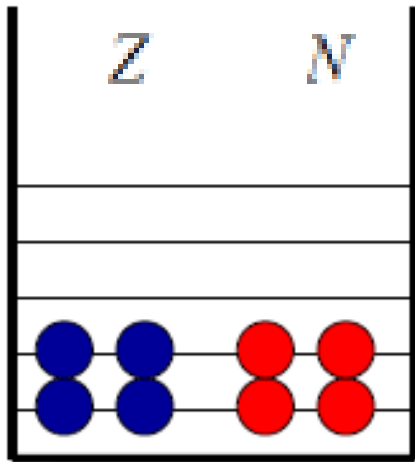
We will assume that the well has approximately equally spaced energy levels.

Let the spacing be ΔE .

Neutrons and protons are fermions (spin $\frac{1}{2}$ particles).

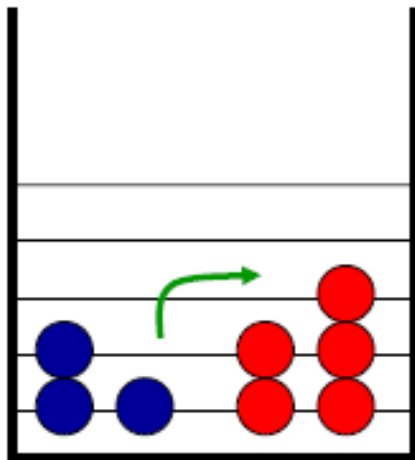
So the Pauli exclusion principle allows only 2 p and 2 n in each level.





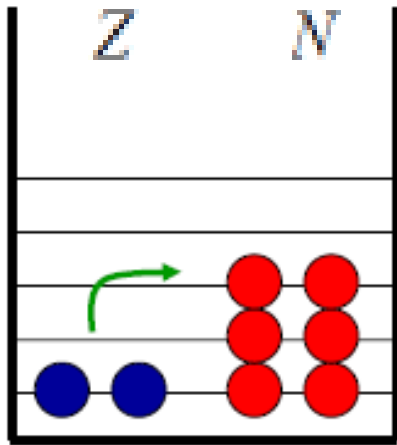
For a nucleus with a given A , *the lowest energy will be when $N = Z$.*

$$(Z - N = 0)$$



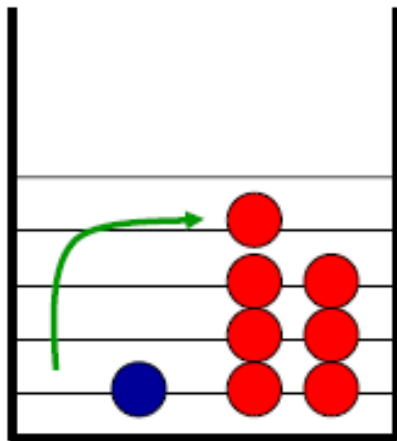
If we change a proton to a neutron we increase the energy by $1\Delta E$ from $Z-N = 0$ case.

$$(\text{Now } Z-N = 2)$$



If we change another proton to a neutron ... we increase the energy by ΔE for a total change of $2\Delta E$ from $Z - N = 0$ case.

(Now $Z - N = 4$)



Another proton to a neutron...
we increase the energy by $3 \Delta E$
for a total change of $5 \Delta E$
from $Z - N = 0$ case.

(Now $Z - N = 6$)

Continuing this, we can find the energy shift for a change in $Z - N$ or $N - Z$.

$Z - N$ or $N - Z$	Total Energy Shift (ΔE)	$\frac{(N - Z)^2}{8}$
2	1	0.5
4	2	2
6	5	4.5
8	8	8
10	13	12.5
12	18	18
14	25	24.5
16	32	32

We can approximate this energy increase by $\frac{(N - Z)^2}{8} \Delta E$

Asymmetry Energy

It has been found that ΔE decreases with A like A^{-1} .
Therefore we can write the asymmetry term as

$$E_a = -a_4 \frac{(N - Z)^2}{A} = -a_4 \frac{(A - 2Z)^2}{A}$$

Pairing Energy

The last correction term arises from the tendency of proton pairs and neutron pairs to occur. Even-even nuclei are the most stable and hence higher binding energies than would otherwise expected. Thus such nuclei appears as peaks in the empirical curve of binding energy per nucleon. At the other extreme, odd- odd nuclei have both unpaired protons and neutrons and have relatively low binding energies.

The pairing energy E_p is positive for even-even nuclei, 0 for odd-even nuclei and even-odd nuclei, and negative for odd-odd nuclei. It seems to vary with A as $A^{-3/4}$. Hence

$$E_p = (\pm, 0) a_5 / A^{3/4}$$

Semi-Empirical Binding Energy Formula

$$E_b = a_1A - a_2A^{2/3} - a_3\frac{Z(Z-1)}{A^{1/3}} \\ - a_4\frac{(A-2Z)^2}{A} (\pm, 0) \frac{a_5}{A^{3/4}}$$

A set of coefficients that gives a good fit with the data is as follows:

$$\begin{array}{lll} a_1 = 14.1 \text{ MeV} & a_2 = 13.0 \text{ MeV} & a_3 = 0.595 \text{ MeV} \\ a_4 = 19.0 \text{ MeV} & a_5 = 33.5 \text{ MeV} & \end{array}$$

Merits of Liquid Drop Model

- The liquid drop model accounts for many of the salient features of nuclear matter, such as the observed binding energies of nuclei and their stability against α and β disintegration as well as nuclear fission.
- The calculation of atomic masses and binding energies can be done with good accuracy with the liquid drop model.

Demerits of Liquid Drop Model

- It fails to explain the extra stability of certain nuclei, where the number of protons or neutrons in the nucleus are 2, 8, 20, 28, 50, 82, or 126.
- It fails to explain the measured magnetic moments of many nuclei
- it also fails to explain the spin of nuclei
- It could not explain the excited states in most of the nuclei.